📘 **AI/ML Internship – Day 14 Notes & Tasks**

🟢 **Module 3: Python for AI**

📌 **Part 1: Working with JSON (Very Important)**

🔹 **What is JSON?**

JSON (JavaScript Object Notation) is a format used to store and exchange data.

👉 Looks like dictionary:

```
{
  "name": "John",
  "marks": 80
}
```

🔹 **Why JSON?**

- Used in APIs
- Used in AI datasets
- Easy to read and write

🔹 **Convert Python → JSON**

```
import json

data = {"name": "Sara", "marks": 90}

json_data = json.dumps(data)
print(json_data)
```

🔹 **Convert JSON → Python**

```
json_data = '{"name": "Sara", "marks": 90}'

data = json.loads(json_data)
print(data["name"])
```

🔹 Write JSON to File

```
import json

data = {"name": "Alex", "marks": 85}

with open("data.json", "w") as file:
    json.dump(data, file)
```

🔹 Read JSON from File

```
import json

with open("data.json", "r") as file:
    data = json.load(file)

print(data)
```

📌 Part 2: Working with CSV Files

🔹 What is CSV?

CSV = Comma Separated Values
Used for storing tabular data.

🔹 Example CSV

```
name,marks
John,80
Sara,90
```

🔹 Read CSV

```
import csv

with open("data.csv", "r") as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

🔹 **Write CSV**

```
import csv

with open("data.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["name", "marks"])
    writer.writerow(["John", 80])
```

📌 **Part 3: Virtual Environment (Important Concept)**

🔹 **What is Virtual Environment?**

A virtual environment is an isolated space for Python projects.

🔹 **Why Use It?**

- Avoid conflicts
- Manage dependencies
- Professional development

🔹 **Create Virtual Environment**

```
python -m venv myenv
```

🔹 **Activate**

```
myenv\Scripts\activate   # Windows
```

📌 **Part 4: Requirements File**

🔹 **What is requirements.txt?**

File that stores project dependencies.

🔹 **Create**

pip freeze > requirements.txt

🔹 **Install**

pip install -r requirements.txt

📌 **Part 5: Real-World Thinking**

👉 In AI/ML:

- Data comes in JSON/CSV
- Projects use virtual environments
- Libraries are managed using requirements

👉 Today connects directly to:

- APIs
- Datasets
- ML workflows

🎯 **Tasks for Day 14**

🧠 **Task 1: Theory**

1. What is JSON?
2. Difference between JSON and dictionary
3. What is CSV file?
4. What is virtual environment?
5. What is requirements.txt?

💻 **Task 2: JSON Practice**

1. Convert dictionary to JSON
2. Convert JSON to dictionary

3. Write JSON to file
4. Read JSON from file

💻 **Task 3: CSV Practice**

1. Create CSV file
2. Write student data
3. Read CSV file
4. Print formatted output

💻 **Task 4: Combined Task**

👉 Program:

- Read CSV file
- Convert to dictionary
- Save as JSON

💻 **Task 5: Environment Practice**

1. Create virtual environment
2. Activate it
3. Install any package
4. Generate requirements.txt